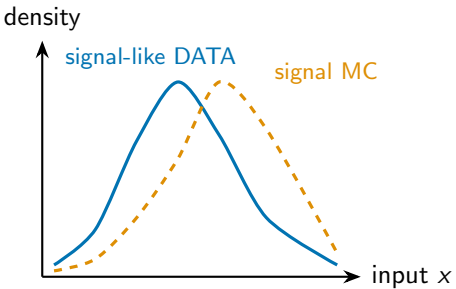
Why reweight the signal simulation?



1. XGBoost learns the reconstructed signal distribution in MC.
2. A DATA–MC mismatch changes the learned decision boundary.
3. The selected signal efficiency then depends on the mismodelled variables.
4. The selected mass spectrum and downstream fit inherit this dependence.

A stronger classifier does not by itself remove signal-domain mismodelling.

Reweighting as a multidimensional density-ratio correction

$$w(\mathbf{x}) \simeq \frac{p_{\text{DATA signal}}(\mathbf{x})}{p_{\text{MC signal}}(\mathbf{x})}$$

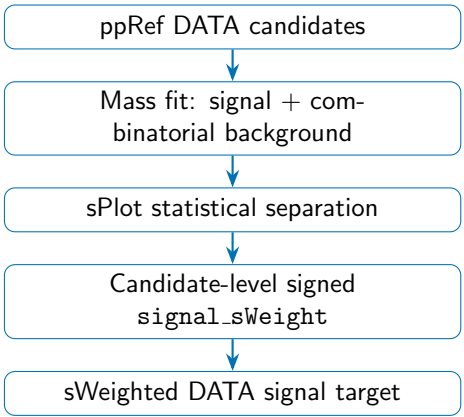
$$p_{\text{MC}}^{\text{weighted}}(\mathbf{x}) \approx p_{\text{DATA signal}}(\mathbf{x}).$$

- $\mathbf{x}$ contains correlated reconstruction-level variables.
- One continuous weight is assigned to each simulated candidate.
- Event reconstruction and measured DATA are unchanged.
- No additional simulated candidates are created.

Original signal MC training population $\xrightarrow{w(\mathbf{x})}$ Weighted signal MC DATA-like training population

BDT reweighting follows the multidimensional approach of A. Rogozhnikov, arXiv:1608.05806.

The DATA target must represent the signal component



- Negative sWeights are retained.
- The target is not an event-by-event truth label.
- Fit quality and signed effective statistics must be checked.
- The correction is defined only inside the common target support.

Inclusive sideband or candidate DATA cannot be used as the signal target.

Why the working strategy uses a direct X target

$\psi(2S)$-derived transfer

- Target: ppRef $\psi(2S)$ signed-sWeighted DATA.
- Test: learn a correction in shared variables and apply it to X MC.
- Boundary: limited X point-closure improvement; robust all-variable closure was not established.

Direct-X target

- Target: ppRef X signed-sPlot signal.
- Test: learn the correction directly for X MC and transfer it inside common ppRef/PbPb support.
- Boundary: selected variables and weight tails are better controlled; independent track-dR closure remains unresolved.

Working route: accepted ppRef X signal target + controlled common phase space.

This choice defines a practical workflow; it does not by itself validate ppRef-to-PbPb transfer.

Validation gates before downstream use

Gate	Question answered
Signal target	Is the mass fit usable, and are the signed sWeight sum and effective statistics controlled?
Weight stability	Are the maximum weight, weighted-MC N_{eff} and fold-to-fold behaviour acceptable?
Point closure	Do per-variable signed empirical-CDF distances improve, including variables not used by the reweighter?
Domain support	Are ppRef and PbPb MC compared in the same phase space in which the correction is applied?
Downstream chain	Are weighted/unweighted ML, mass-fit stability and matched-model signal recovery checked separately?

- Passing a domain-classifier or aggregate closure metric is not sufficient on its own.
- Matched-model injection checks workflow sensitivity, not DATA–MC transfer validity.